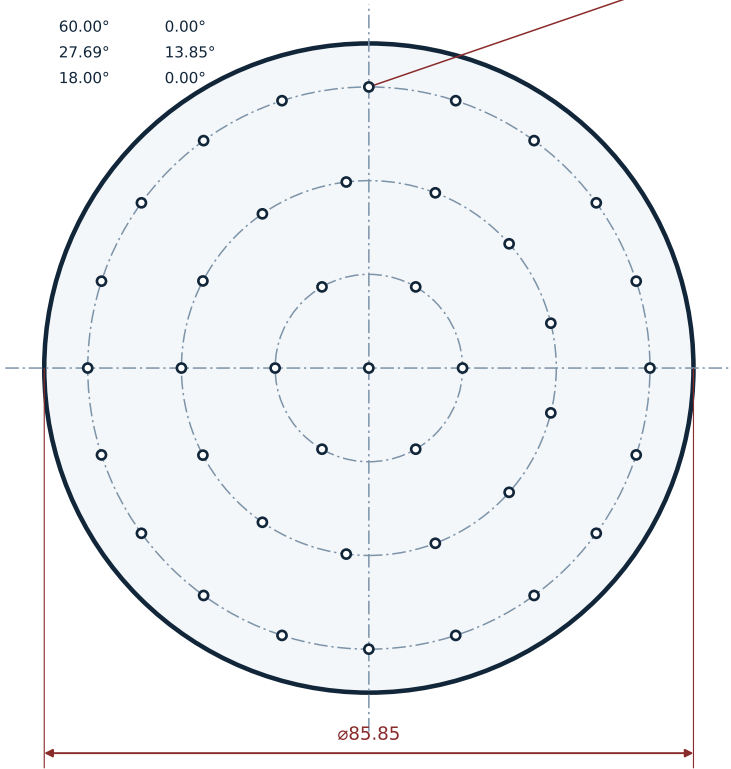


RING DATA

RING	B.C. ø	HOLES	PITCH	STAGGER
C	on centre	1		
1	24.78	6	60.00°	0.00°
2	49.57	13	27.69°	13.85°
3	74.35	20	18.00°	0.00°

40x ø1.200 THRU
L/D 10.58 AREA 45.24 mm²



VIEW ON INLET (UPSTREAM) FACE

EDGE VIEW — HOLES PROJECTED, ALL THRU



12.70 THK

HOLE TABLE

mm from plate centre, X right / Y up

NO	RING	X	Y	R	ANG
1	C	+0.000	+0.000	0.000	0.00°
2	1	+12.392	+0.000	12.392	0.00°
3	1	+6.196	+10.731	12.392	60.00°
4	1	-6.196	+10.731	12.392	120.00°
5	1	-12.392	+0.000	12.392	180.00°
6	1	-6.196	-10.731	12.392	240.00°
7	1	+6.196	-10.731	12.392	300.00°
8	2	+24.063	+5.931	24.783	13.85°
9	2	+18.551	+16.434	24.783	41.54°
10	2	+8.788	+23.173	24.783	69.23°
11	2	-2.987	+24.603	24.783	96.92°
12	2	-14.078	+20.396	24.783	124.62°
13	2	-21.944	+11.517	24.783	152.31°
14	2	-24.783	+0.000	24.783	180.00°
15	2	-21.944	-11.517	24.783	207.69°
16	2	-14.078	-20.396	24.783	235.38°
17	2	-2.987	-24.603	24.783	263.08°
18	2	+8.788	-23.173	24.783	290.77°
19	2	+18.551	-16.434	24.783	318.46°
20	2	+24.063	-5.931	24.783	346.15°
21	3	+37.175	+0.000	37.175	0.00°
22	3	+35.355	+11.488	37.175	18.00°
23	3	+30.075	+21.851	37.175	36.00°
24	3	+21.851	+30.075	37.175	54.00°
25	3	+11.488	+35.355	37.175	72.00°
26	3	+0.000	+37.175	37.175	90.00°
27	3	-11.488	+35.355	37.175	108.00°
28	3	-21.851	+30.075	37.175	126.00°
29	3	-30.075	+21.851	37.175	144.00°
30	3	-35.355	+11.488	37.175	162.00°
31	3	-37.175	+0.000	37.175	180.00°
32	3	-35.355	-11.488	37.175	198.00°
33	3	-30.075	-21.851	37.175	216.00°
34	3	-21.851	-30.075	37.175	234.00°
35	3	-11.488	-35.355	37.175	252.00°
36	3	-0.000	-37.175	37.175	270.00°
37	3	+11.488	-35.355	37.175	288.00°

... 3 more — export the hole table CSV

NOTES

- 1. ALL DIMENSIONS IN MILLIMETRES. ORIGIN AT PLATE CENTRE. PLAN VIEW IS FROM THE INLET (UPSTREAM) FACE.
- 2. 40 HOLES ø1.200 THRU, EQUALLY SPACED ON THE BOLT CIRCLES LISTED IN RING DATA.
- 3. FLOW AREA GOES AS d²: A HOLE ø ERROR OF +0.02 GIVES +3.4% AREA, AND THE SAME SHIFT IN OXIDISER FLOW. SET THE TOLERANCE TO SUIT.
- 4. Cd 0.700 ASSUMED FOR SHARP-EDGED DRILLED HOLES. DEBURR THE OUTLET FACE. DO NOT CHAMFER OR RADIUS THE INLET WITHOUT RE-SIZING — INLET GEOMETRY CHANGES Cd.
- 5. L/D 10.58 FROM 12.70 THICKNESS. THE FLOW MODEL ASSUMES THIS LENGTH; CHANGING THICKNESS CHANGES PREDICTED FLOW.
- 6. MIN WEB BETWEEN HOLES 10.43. EDGE MARGIN 5.15.
- 7. PRINT AT 100% ON A4 LANDSCAPE FOR TRUE SCALE — CONFIRM WITH THE CHECK BAR BEFORE MEASURING OFF THIS SHEET.
- 8. SIZED ON CHAMBER PRESSURE 25 bar. SEE TRADE STUDY AND CONFIG NOTES.

CHECK BAR — 50 mm ON PAPER:



INJECTOR ORIFICE PLATE

PROJECT	Mini Hybrid V2.2
PART NO	INJ-PLATE-V2.2C
MATERIAL	—
PLATE	ø85.85 × 12.70 THK
HOLES	40 × ø1.200 THRU
SCALE	1 : 1 UNITS mm
DRAWN	Molen 2026-09-26
HASTE v1.0.0 — HEM model, Cd 0.700	